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AUTOMATED INTERACTIVE VISUALIZATION ON ABSTRACT CONCEPTS IN COMPUTER SCIENCE

OUTLINE

- Introduction
- Design methodology
- Implementation
- Experimental results
  - Offline course experiment
  - Online course experiment
- Conclusion and future works

INTRODUCTION(1)

- **Goals**
  - Resolve difficulties of teaching computer science concepts
  - Resolve difficulties of make interactive visualizations of computer science concepts

INTRODUCTION(2)

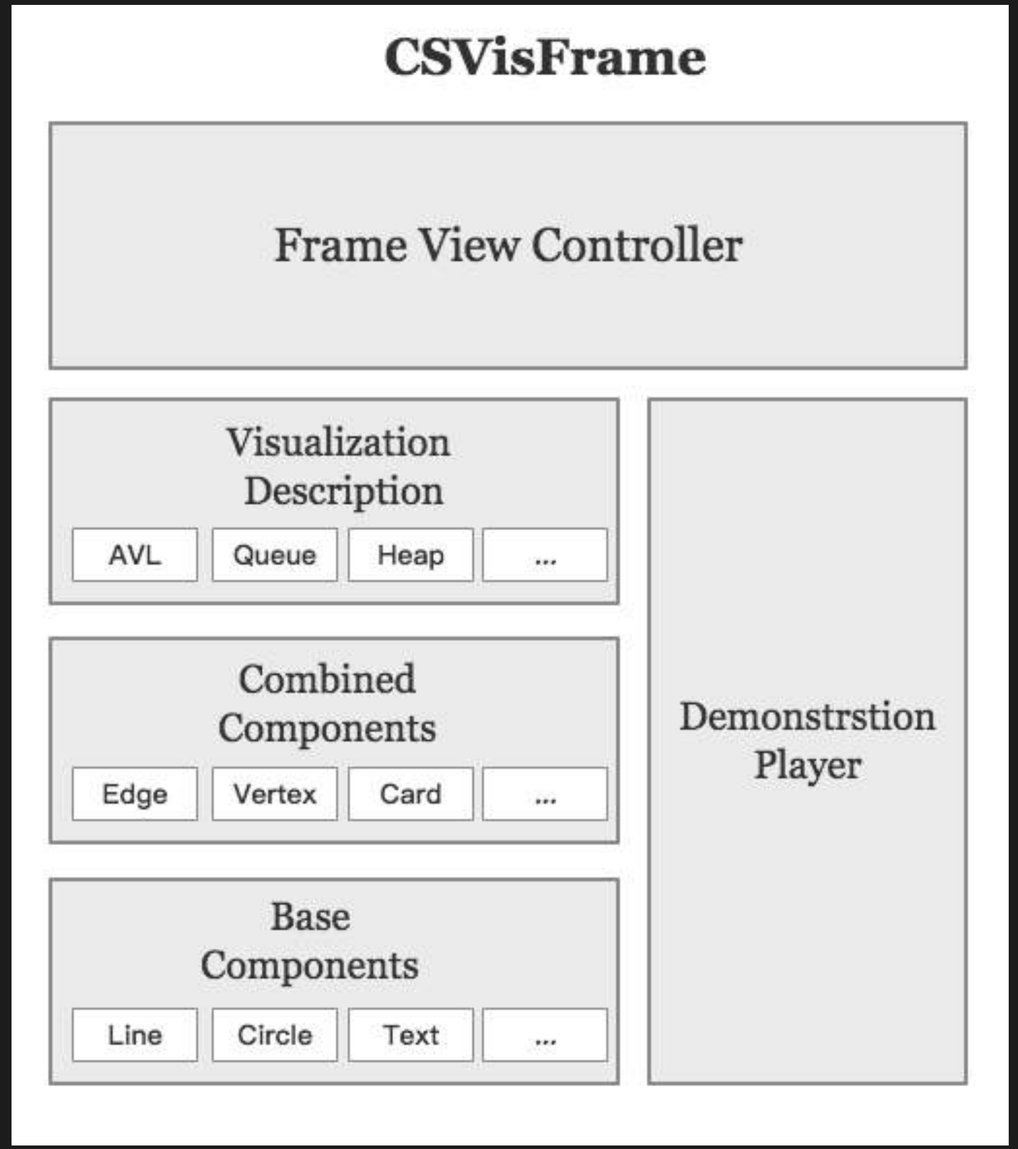
- **Main contributions**
  - CSVisFrame: a framework for making visualizations of computer science concepts
  - Test result of learning effectiveness affected by CSVisFrame based interactive visualizations

DESIGN METHODOLOGY

- We want to build a framework which is in common use.
- Four considerations below:
  - Hierarchical
  - Cross-platform
  - Interactive support
  - Language Independent

IMPLEMENTATION(1)——HIERARCHICAL DESIGN

- Four layers of hierarchic
- Demonstration player
- Frame View Controller

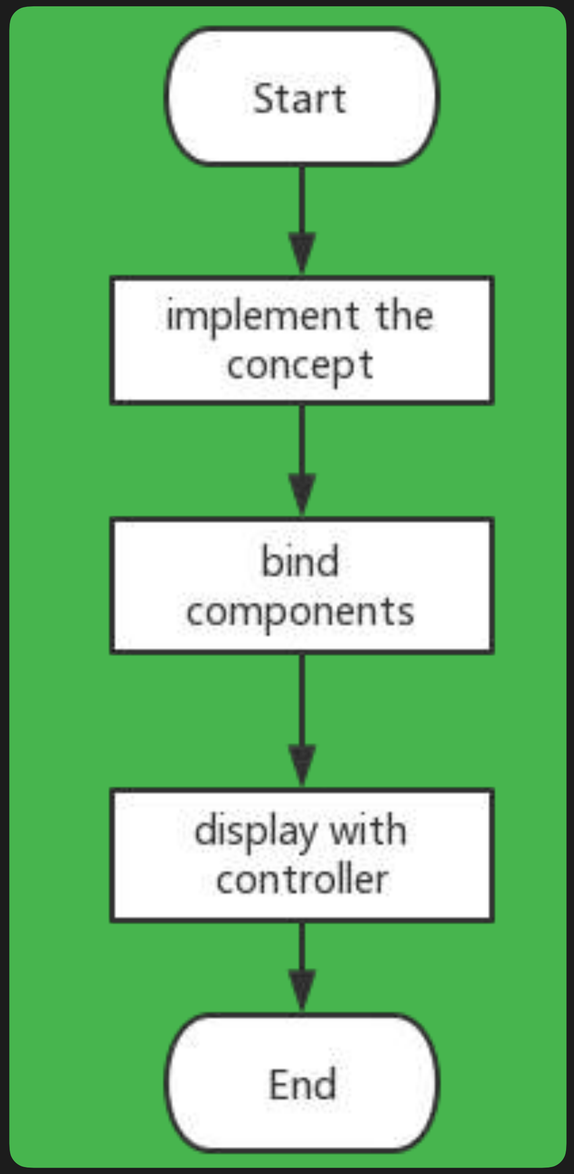


IMPLEMENTATION(2)——EXTENDABLE STRUCTURE

- CSVisFrame is extendable on lower three layers
  - Base components
  - Combined components
  - Visualization descriptions

IMPLEMENTATIONS(3)—VISUALIZATION WITH CSVISFRAME

- Implement relevant concepts and logics
- bind components to the data structure
- generate the buttons and input boxes to display under control



(A)

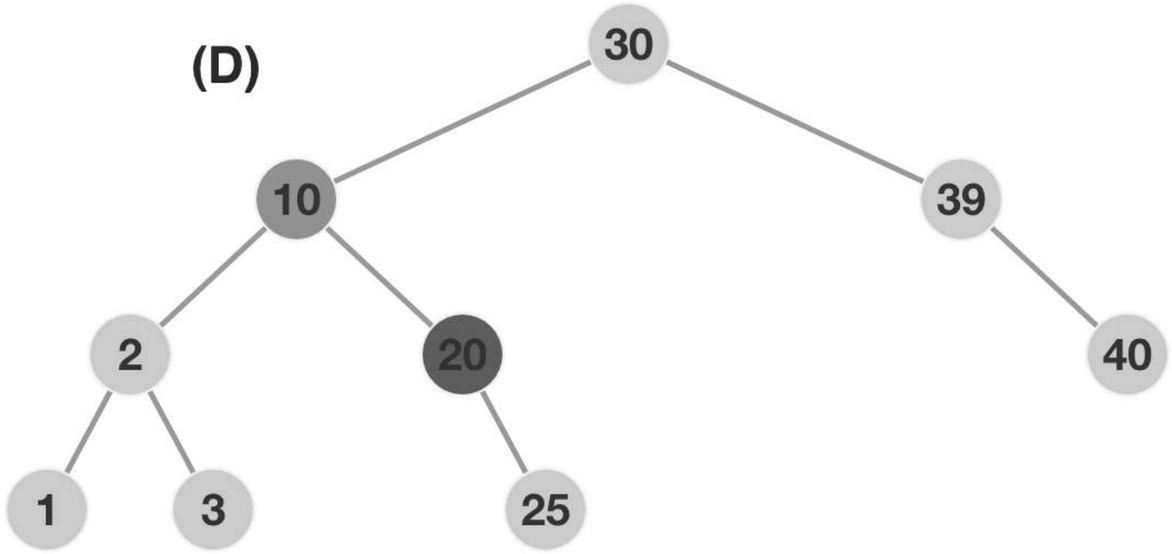
插入 删除 查找

(B)

(C)

Right-Rotate 节点 20

(D)



(E)

```
1 function RightRotate(T)
2     temp = T.left
3     T.left = temp.right
4     temp.right = T
5     updateHeight(T)
6     updateHeight(temp)
7     return temp
```

(F)

完成互动

IMPLEMENTATION(4)——EXAMPLE: AVL TREE

- lots of steps…

OFFLINE COURSE EXPERIMENT

- *Algorithm Design and Analysis* in Sichuan Normal University
- We teach them Binary Search Tree and AVL Tree in the classroom
- 60+ sophomores

ONLINE COURSE EXPERIMENT

- Data structure on Jisuanke Platform: https://www.jisuanke.com/course/cs261
- Queue, Stack, Heap, Size Balanced Tree, Prim Algorithm and other demonstrations are implemented with CSVisFrame



EXPERIMENT RESULT(1)

TABLE I. Survey Questions

Q1	Visualization is more intuitive for me
Q2	I think interactive visualization is better than animation
Q3	Visualization helps my understanding
Q4	I understand the data structure better after using the visualization
Q5	The visualization is hard to use and worthless
Q6	How you judge the visualization compare to traditional methods?

TABLE II. Result from students of Sichuan Normal University

Question	Average Score
Q1	6.35
Q2	6.52
Q3	6.18
Q4	5.69
Q5	1.92

TABLE III. Result from online students studying on Jisuanke

Question	Average Score
Q1	6.28
Q2	6.28
Q3	6.07
Q4	5.61
Q5	1.46

EXPERIMENT RESULT(2)——STUDENT COMMENTS

- The single step mode is very useful. Control it according to my understanding ability is preferable to play the animation directly.
- It could be better if shortcut on the keyboard can be added to control the visualization demonstration.
- The colour in the visualization could be enhanced for better visual experience.

CONCLUSION

- Visualizations can be created based on CSVisFrame.
- Visualizations made by CSVisFrame are effective in helping students obtaining abstract concepts in particular fields.

FUTURE WORKS

- Add shortcuts on the keyboard
- Support more context information about current instruction
- Open source of the CSVisFrame(done)
  - github.com/Jisuanke/CSVisFrame

THANKS FOR YOUR ATTENTION